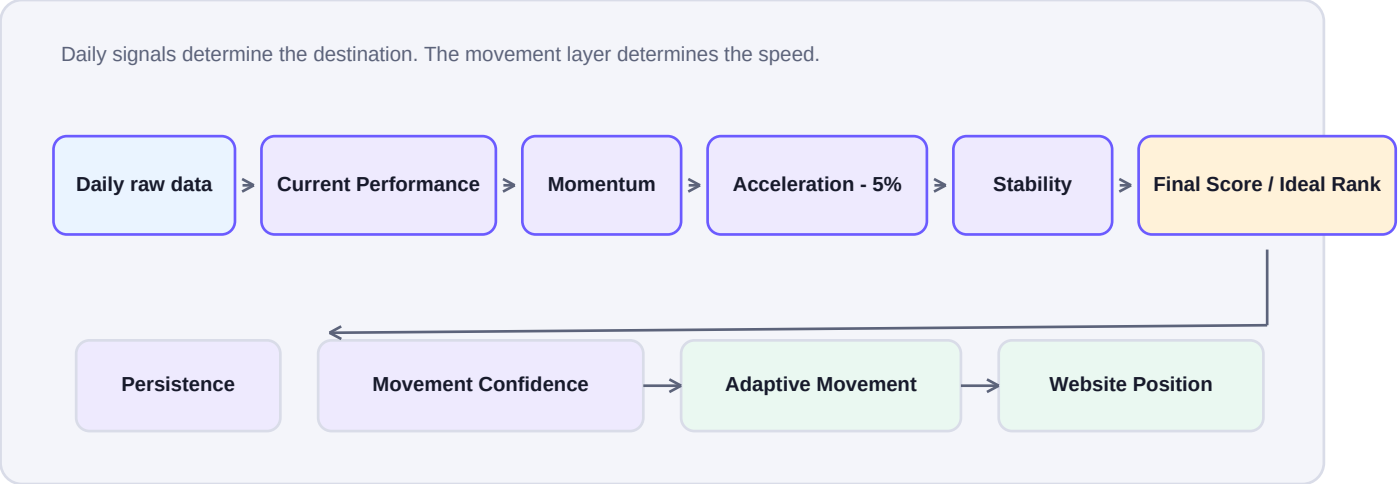
CATEGORY X - RANKING ENGINE

How the Ranking Algorithm Works

v14 detailed calculation guide: signals, score, persistence, movement confidence and adaptive website placement



Purpose. The engine ranks games every day. Daily performance signals determine an ideal rank. A separate movement layer controls how quickly the actual website position moves toward that ideal rank. This separation is important because a game can deserve a better position without needing to jump there in one update.

Core component	Weight	Main question
Current Performance	65%	How strong is the game now?
Momentum	15%	Is it improving or declining?
Acceleration	5%	Is the trend strengthening or weakening?
Stability	15%	How consistent is the recent pattern?
Persistence	Movement multiplier	How long has the direction continued?
Movement Confidence	Movement layer	How much evidence supports moving this game?

1. The complete architecture

The algorithm has two stages. The first stage determines where the game ideally belongs. The second stage decides how quickly the website position should move toward that ideal rank.

Stage	Calculation	Output
A - Ranking signal calculation	Current Performance + Momentum + Acceleration + Stability	Final Score and Ideal Rank
B - Movement control	Evidence + Momentum + Acceleration + Persistence + incumbent protection	Actual Website Position

The key distinction is Final Score versus Website Position. A final score of 90 does not mean the game instantly becomes position #10. The score creates the ideal destination, while the movement engine controls the path used to get there.

2. Current Performance - 65%

Current Performance is the main signal because it captures the game's recent actual strength. It uses the most recent 7 days for the game, with newer observations given greater influence than older observations. Four raw performance measures are combined: Clients, Bet Count, Stakes and GGR.

Metric	Weight inside Current Performance	Reason
Clients	20%	Shows breadth of recent player activity.
Bet Count	15%	Shows recent betting activity.
Stakes	30%	Shows the value of betting activity.
GGR	35%	Shows recent gross gaming revenue contribution.

The four values are normalized to a comparable 0-100 scale and then combined.

Worked example. Suppose the latest 7-day calculation gives normalized values of Clients = 96, Bet Count = 99, Stakes = 99 and GGR = 98.

Current Performance = (96 x 0.20) + (99 x 0.15) + (99 x 0.30) + (98 x 0.35) = 19.20 + 14.85 + 29.70 + 34.30 = 98.05.

3. Momentum - 15%

Momentum measures direction and strength. It compares recent activity with the preceding activity over three windows: 3-day, 7-day and 14-day. The 3-day comparison is more responsive, the 7-day comparison provides the main recent trend, and the 14-day comparison gives longer context.

Window	Role	Interpretation
3-day	Fast signal	Detects a recent change quickly.
7-day	Main trend	Balances responsiveness with noise control.
14-day	Longer context	Checks whether the recent change fits a broader pattern.

Momentum is signed. Positive momentum means the game is improving. Negative momentum means the game is declining. The multi-window design prevents the engine from relying on one short burst of activity.

Example: 3-day momentum = +40, 7-day momentum = +25 and 14-day momentum = +10. All three are positive, so the game has an improving trend. The much stronger 3-day result also tells us the improvement is more recent than the longer history suggests.

4. Acceleration - 5%

Acceleration measures how Momentum itself is changing. It is not the same as Momentum. Momentum answers "which direction?" while Acceleration answers "is that direction getting stronger or weaker?"

Momentum	Acceleration	Meaning
+30	+20	Improving, and the improvement is strengthening.
+30	-20	Still improving, but the improvement is cooling.
-30	+20	Still declining, but the decline is slowing/recovering.
-30	-20	Declining, and the decline is becoming stronger.

Worked example: Momentum moves from -30 to -10. The change is +20, so Acceleration is positive. The game is still declining because Momentum remains below zero, but its decline is becoming less severe.

Acceleration is kept at 5% in v14. Our sensitivity testing showed that larger weights increased movement and reduced Top-10/Top-20 overlap on the current evaluation data. The 5% weight therefore gives acceleration influence without allowing it to dominate the stronger Current Performance signal.

5. Stability - 15%

In v14, Stability is a consistency signal. It looks at whether recent performance follows a repeatable pattern or contains large swings. High Stability means the recent pattern is more consistent; low Stability means it is more volatile.

Important: Stability is not a direct instruction to keep the game in the same website position. A game can move rapidly while still being stable if the movement is supported by consistent performance evidence.

Performance pattern	Expected Stability	Why
100, 110, 120, 130, 140...	High	Large change, but smooth and consistent direction.
100, 160, 90, 170, 80...	Low	Large swings and reversals make the pattern volatile.

The v14 definition was later reconsidered during development. v15 attempted a different Stability formulation; that version introduced behaviour the project did not want, so v14 remains the working baseline. This guide therefore documents v14, not the v15 experiment.

6. Final Score - the ideal destination

The final score combines the four ranking signals using the v14 weights:

Component	Value	Weight	Contribution
Current Performance	90	65%	58.50
Momentum	70	15%	10.50
Acceleration	60	5%	3.00

Component	Value	Weight	Contribution
Stability	80	15%	12.00
Final Score		100%	**84.00**

The games are then sorted by Final Score to establish the **Ideal Rank**. Ideal Rank is not necessarily the same as the current website position.

7. Direction Persistence

Direction Persistence measures how many consecutive ranking days support the current direction. If Momentum remains positive across consecutive days, the positive persistence streak grows. If Momentum changes direction, the streak resets and a new streak begins.

Day	Momentum	Direction	Persistence
1	+18	Improving	1 day
2	+25	Improving	2 days
3	+31	Improving	3 days
4	+36	Improving	4 days
5	-8	Declining	1 day declining streak

Persistence is not a waiting-period rule. A game does not have to wait a fixed number of days before it can overtake another game. Persistence is a confidence multiplier: when the evidence keeps pointing in the same direction, more of the movement pressure is allowed to affect the website position.

8. Movement Confidence

Movement Confidence is a movement-layer concept. It represents how strongly the available evidence supports changing the game's current website position. It considers the strength of the signals and the persistence of the direction, rather than simply asking whether the game moved yesterday.

Evidence pattern	Movement confidence	Expected behaviour
Weak/mixed Momentum, low persistence	Low	Hold or make a small change.
Strong Momentum + positive Acceleration, 1-2 days	Moderate	Allow visible but controlled movement.
Strong Momentum + positive Acceleration + 4+ days	High	Allow stronger movement toward Ideal Rank.
Evidence turns negative	Falls	Upward pressure reduces; downward movement can build if decline persists.

The exact movement-confidence value shown in the UI is therefore best understood as a **decision-support signal**, not as a separate performance score. It helps explain why one game is allowed to move faster than another.

9. Adaptive Movement

Adaptive movement is the mechanism that turns Ideal Rank into the actual Website Position. Instead of applying one fixed number of positions to every game, v14 varies movement according to evidence strength,

Momentum/Acceleration and persistence. Strong established games also receive protection from abrupt downward replacement.

Input	Role in movement
Distance to Ideal Rank	Shows how far the game needs to travel.
Momentum	Creates directional pressure.
Acceleration	Strengthens or weakens that directional pressure.
Persistence	Multiplies how much of the pressure is trusted.
Stability / evidence	Supports confidence and reduces overreaction to weak/noisy evidence.
Incumbent protection	Makes established strong games harder to displace abruptly.

Conceptual rule: base movement pressure x persistence multiplier, subject to adaptive caps and incumbent protection. The important point is that movement is not based on persistence alone.

10. Full end-to-end calculation example

The example below is illustrative. It shows how the signals and movement logic fit together using the v14 architecture. The individual example values are chosen for explanation and are not a claim about one particular real game.

Signal	Example value	What it says
Current Performance	88	Strong recent 7-day performance.
Momentum	+42	Strong upward trend.
Acceleration	+24	The upward trend is strengthening.
Stability	82	Recent pattern is reasonably consistent.
Persistence	4 days	The positive direction has persisted.
Current website position	#200	Game is currently far below its likely ideal position.
Ideal rank	#70	Final Score indicates a substantially better target position.

Step 1 - Final Score. Using the v14 weights: (88 x 0.65) + (42 x 0.15) + (24 x 0.05) + (82 x 0.15) = 57.20 + 6.30 + 1.20 + 12.30 = **77.00**.

The score of 77.00 determines the game's relative Ideal Rank. Suppose this places the game at Ideal Rank #70.

Step 2 - Direction. Momentum is +42 and Acceleration is +24. Both support upward movement. Persistence is already 4 days, so the evidence is not a single-day spike.

Step 3 - Movement confidence. Strong Momentum + positive Acceleration + 4-day persistence creates high movement confidence. The game therefore receives a larger upward movement allowance than a similar game with only one day of positive evidence.

Step 4 - Adaptive move. The game does not jump from #200 directly to #70. The movement layer chooses a controlled step. For illustration only, suppose the calculated allowance for this evidence is 22 positions.

Day 1: #200 -> #178. If the evidence remains strong, persistence increases and the next allowed move can become larger.

Day 2: #178 -> #155. Momentum and Acceleration remain strong; persistence is now 5 days.

Day 3: #155 -> #128. The game keeps progressing toward #70 because the evidence remains strong.

Day 4: #128 -> #103. The game is moving quickly, but still progressively. It is not being teleported to its ideal rank.

Why this matters: this is the core business behaviour we want. A genuine breakout can become visible, while weak or short-lived changes are less likely to cause a dramatic position swap.

11. Established game versus new challenger

The ranking engine must balance two competing needs: protect established strong games from temporary spikes, while still allowing a genuinely superior challenger to overtake them.

	Established game	Challenger
Website position	#20	#45
Current Performance	82	88
Momentum	+15	+42

	Established game	Challenger
Acceleration	+3	+24
Stability	88	75
Persistence	7 days strong	2 days improving

The challenger has better current signals, but its evidence is newer. The established game has a stronger history. The persistence-aware movement logic therefore lets the challenger close the gap while protecting the incumbent from an instant replacement.

If the challenger continues to outperform, its persistence grows and the movement pressure increases. If its performance fades after two days, the upward pressure disappears and the challenger does not keep the benefit indefinitely.

12. Missing data behaviour

Missing data is not treated as zero performance. If a game has no selected-day record, the engine labels it No Data, keeps its last website position, and does not invent a new ranking signal from zeros.

Data state	Raw metrics	Website position
Data available	Calculated normally	Can move according to evidence
No data	No Data / not zero	Held from prior position

The point is to distinguish "missing information" from "bad performance." A missing row should not push a game down as if it had zero Clients, zero Bets, zero Stakes and zero GGR.

13. What we learned from testing

Several tests were used during v14 development. The most important findings were behavioural rather than just numerical.

Test	Finding	Implication
Acceleration sensitivity	Weights above 5% reduced Top-10/Top-20 overlap in the tested data.	Keep Acceleration at 5% for the v14 baseline.
Persistence movement	Large movements and reversals were reduced while Top-20/Top-100 quality stayed close to baseline.	Persistence is useful as a movement multiplier.
Overtake scenarios	Strong/persistent challengers could overtake; weak or fading challengers were constrained.	Evidence matters more than a fixed waiting period.
Real-data A/B movement	Persistence made movement much calmer and improved Top-10 overlap in the tested period.	The architecture is promising, but breakout responsiveness must be monitored.

14. Practical interpretation of the Dashboard

UI metric	Plain-English meaning
Current Performance	How strong the game has been recently.
Momentum	Whether the game is moving up or down and how strongly.

UI metric	Plain-English meaning
Acceleration	Whether that improvement/decline is speeding up or slowing down.
Stability	How consistent the recent performance pattern is.
Direction Persistence	How long the current improving/declining direction has continued.
Movement Confidence	How much the combined evidence supports changing the website position.
Website Position	The actual simulated position shown to users.
Ideal Rank	Where the final score says the game ideally belongs.

One sentence summary: Current Performance tells us how strong a game is, Momentum tells us where it is going, Acceleration tells us whether that trend is strengthening, Stability tells us how consistent the pattern is, Persistence tells us how long the evidence has lasted, and Movement Confidence helps convert that evidence into a sensible amount of website movement.

This separation is important for the project because the business goal is not to maximize movement. The goal is to let strong, persistent improvements become visible while avoiding rapid, unjustified switching between games.

15. Notes and boundaries

1. All worked examples in this guide are illustrative unless explicitly identified as a v14 test result. They explain the mechanics and are not predictions of future game behaviour.

2. v14 is the project baseline documented here. The later v15 Stability experiment is intentionally excluded because the project reverted to v14.

3. Persistence is not a fixed takeover timer. There is no rule that says a challenger must wait a particular number of days before overtaking. The movement decision is evidence-driven.

4. The exact day-by-day website position depends on the full catalogue because games compete for unique positions. Therefore, examples of exact position changes are illustrative unless generated by a specific pipeline run.